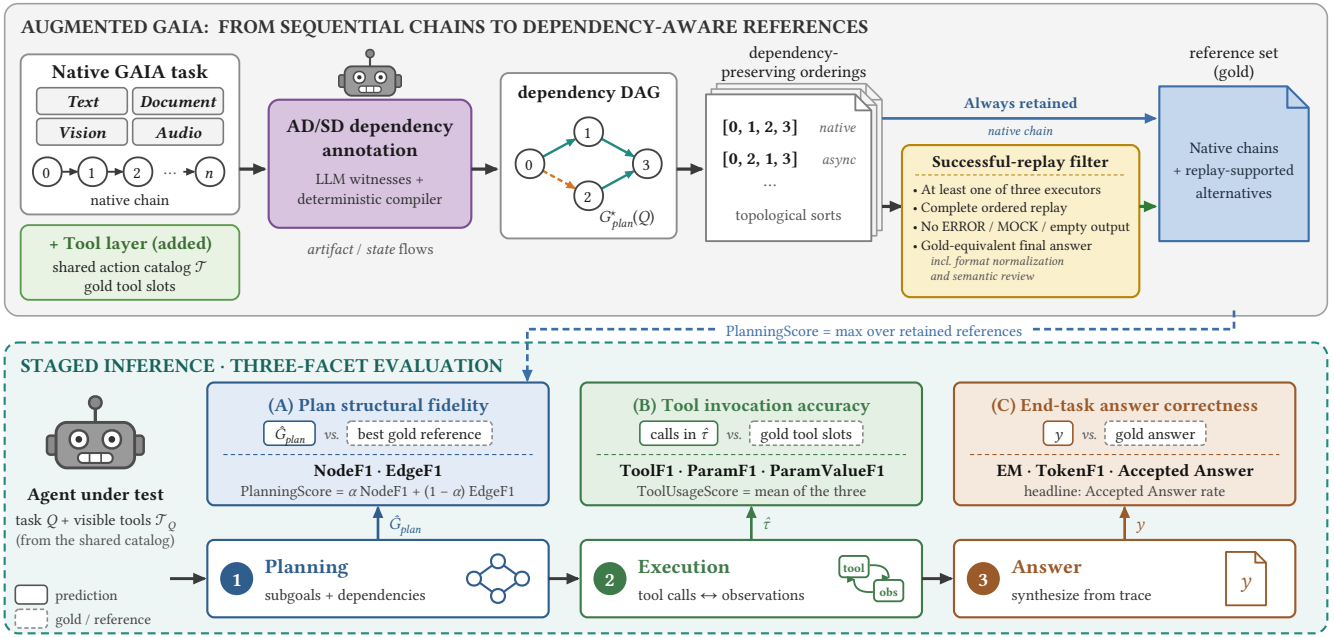

AUGMENTED GAIA: FROM SEQUENTIAL CHAINS TO DEPENDENCY-AWARE REFERENCES
Native GAIA task
Text
Document
Vision
Audio
0
1
2
…
n
native chain
+ Tool layer (added)
shared action catalog $\mathcal{T}$
gold tool slots
AD/SD dependency annotation
LLM witnesses + deterministic compiler
artifact / state flows
dependency DAG
0
1
2
3
$G^{\star}_{plan}(Q)$
dependency-preserving orderings
[0, 1, 2, 3] native
[0, 2, 1, 3] async
…
topological sorts
Always retained
native chain
Successful-replay filter
• At least one of three executors
• Complete ordered replay
• No ERROR / MOCK / empty output
• Gold-equivalent final answer
incl. format normalization and semantic review
reference set (gold)
Native chains + replay-supported alternatives
PlanningScore = max over retained references
STAGED INFERENCE · THREE-FACET EVALUATION
Agent under test
task Q + visible tools $\mathcal{T}_Q$ (from the shared catalog)
prediction
gold / reference
(A) Plan structural fidelity
$\hat{G}_{plan}$ vs. best gold reference
NodeF1 · EdgeF1
PlanningScore = α NodeF1 + (1 − α) EdgeF1
(B) Tool invocation accuracy
calls in $\hat{\tau}$ vs. gold tool slots
ToolF1 · ParamF1 · ParamValueF1
ToolUsageScore = mean of the three
(C) End-task answer correctness
y vs. gold answer
EM · TokenF1 · Accepted Answer
headline: Accepted Answer rate
$\hat{G}_{plan}$
$\hat{\tau}$
y
1 Planning
subgoals + dependencies
2 Execution
tool calls ↔ observations
tool
obs
3 Answer
synthesize from trace
y